- Biotin is also indicated for healthy hair and skin, healthy sweat glands, nerve tissue, and bone marrow.
- Biotin is also helps in maintaining a steady blood sugar level.

Food sources

Biotin is widely distributed in both animal and plant foods. Liver, kidney, egg, yolk, milk, tomatoes are rich sources.

Deficiency

Dietary deficiency of biotin is rare. Symptoms of biotin deficiency include lack of appetite, nausea, an enlarged tongue, mental depression, pallor, loss of hair.

Pantothenic acid (vitamin B6)

Pantothenic acid referred to as the "anti – stress vitamin" is part of the B group vitamins. This vitamin can be produced in the body by the intestinal flora.

Functions

- Pantothenic acid plays an important role in the secretion of hormones, such as cortisone because of the role it plays in supporting the adrenal gland.
- Pantothenic acid is also used in the release of energy as well as the metabolism of fat, protein and carbohydrates.
- It is used in the synthesis of lipids, neurotransmitters and haemoglobin.

Food sources

Beef, brewer's yeast, egg, fresh vegetables, kidney, legumes, liver, mushrooms, nuts, pork, royal jelly, saltwater fish, torula yeast, whole rye flour, and whole wheat contain this vitamin.

Deficiency

Symptoms of a pantothenic acid deficiency though rare, may occur in severely malnourished individuals and include fatigue, irritability, low blood pressure upon standing, lack of appetite, constipation, and tingling and numbness in both the feet and hands.

5.9 Minerals

The essential minerals are classified according to their recommended intake. Macro minerals, also called major minerals, are needed in amounts greater than 100 milligrams per day. Micro minerals, also called trace minerals or trace elements, are needed in less than 100 milligrams per day.

Table: 10 Classification of essential minerals

Macro minerals (> 100 mg/day)*	Micro minerals (<100 mg/day)*
Calcium	Iron
Phosphorus	Zinc
Magnesium	Copper
Sulfur	Iodine
Sodium*	Fluoride
Potassium*	Manganese
Chloride*	Selenium
	Chromium
	Molybdenum

*Recommended intakes are greater or lesser than 100mg per day, as indicated.

Macro minerals

There are seven macro minerals; calcium, phosphorus, magnesium, sulfur, sodium, potassium, and chloride. As well as being needed in amounts greater than 100 milligrams per day, each of the macro minerals makes up more than 0.01% of the body's weight.

Calcium

Calcium is by far the most prevalent mineral in the body. About 98% of the 1200 grams (2.5 pounds) of calcium in the average adult body is found in the bones. Small amounts of calcium (1%) are also found in the extracellular fluid, certain intracellular structures, and cell membranes.

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